Impact Statement: Optimising RAG Document Retrieval for Agronomic Advice

Purpose. This project develops the retrieval engine at the heart of a Retrieval-Augmented Generation system for agricultural extension advice. Given a smallholder farmer's natural-language question — for example, "Why are my maize leaves turning yellow?" — it ranks and returns the most relevant factsheets from a knowledge base covering crop diseases, pests, nutrient deficiencies, soil management, fertiliser use, and climate adaptation. It addresses a concrete community problem: reliable agricultural guidance exists but is scattered and phrased in technical language that rarely matches how farmers ask, so the right advice is hard to surface at the moment it is needed.

Positive impacts. Because the quality of any RAG answer is capped by what retrieval surfaces, improving this step raises the reliability of every downstream answer a farmer receives. Better retrieval means faster, more accurate diagnoses, less wasted input spending, and stronger resilience to pests and climate shocks — benefits that flow to smallholder farmers, the extension workers who advise them, and the food security of their wider communities. Keeping guidance traceable to sourced factsheets also strengthens trust and verifiability.

Potential risks and harms. The most significant risk is confidently retrieving a document that looks relevant but is wrong — surfacing potassium-deficiency advice for a nitrogen problem — which could cause real crop and financial loss. Further risks include uneven quality across crops and regions (over-serving major staples like maize while neglecting minor crops), automation bias where farmers or advisers over-trust system output, and bias inherited from a corpus written largely in standard English rather than local languages or informal phrasing.

Values. The work is guided by accuracy, relevance, inclusivity, fairness, and transparency. These are not abstract: farmers act on what they receive, so relevance and accuracy directly protect livelihoods, inclusivity ensures short and conversational questions are served, and fairness guards against systematically underserving any farming group.

Mitigation strategies. We evaluate against expert relevance judgements using nDCG@5, deliberately test against hard negatives that punish surface keyword matching, and measure retrieval quality separately across crops and query types to detect uneven performance. We treat the system as a decision aid that surfaces sourced documents rather than a replacement for expert judgement, keeping every result traceable to its origin.

Limitations and uncertainties. Our assessment assumes the dataset reasonably reflects real farmer questions; in practice, live queries in local languages may differ substantially, and performance on a small self-contained corpus may not generalise to larger, noisier real-world collections. We also cannot fully anticipate how farmers will interpret or act on surfaced guidance. These uncertainties point to clear next steps: validation with real users and localisation into under-represented languages and crops.